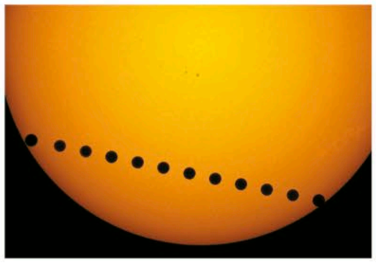
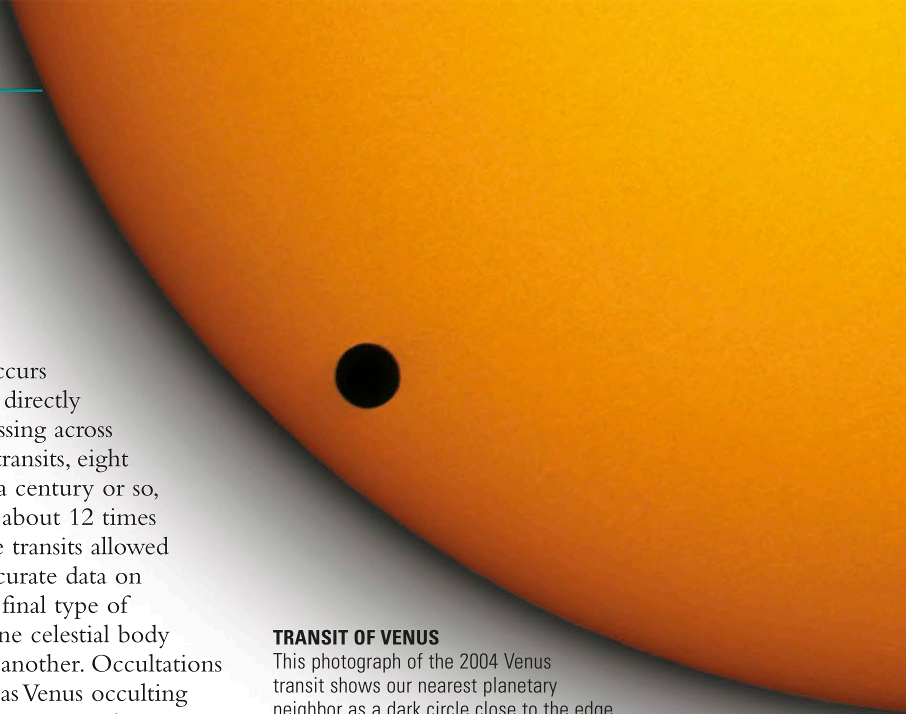
ALIGNMENTS IN THE SKY

Because all the planets orbit the Sun roughly in the same plane (see pp.102–103), they never stray from the band in the sky called the zodiac (see p.65). It is not uncommon for several of the planets to be in the same part of the sky at the same time, often arranged roughly in a line. Such events, called planetary conjunctions, are of no deep significance, but can be a spectacular sight. Another type of alignment, called a transit, occurs when an inferior planet comes directly between Earth and the Sun, passing across the Sun's disk. A pair of Venus transits, eight years apart, occurs about once a century or so, while Mercury transits happen about 12 times a century. In earlier times, these transits allowed astronomers to obtain more accurate data on distances in the solar system. A final type of alignment is an occultation—one celestial body passing in front of, and hiding, another. Occultations of one planet by another, such as Venus occulting Jupiter, occur only a few times a century; in contrast, occultations of one or other of the bright planets by the Moon occur 10 or 11 times a year.



VENUS'S PATH ACROSS THE SUN'S DISK
This composite photograph of Venus's 2004 transit spans just over five hours. During this time, astronomers gathered data on the Sun's changing light to use as a model to look for Earth-sized planets orbiting other stars.



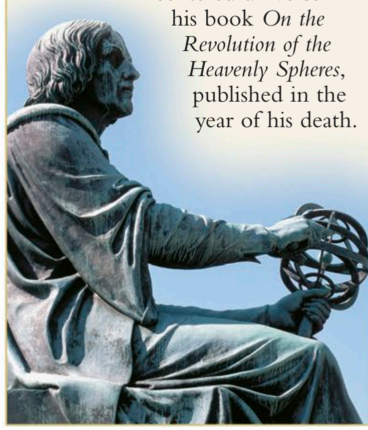
TRANSIT OF VENUS
This photograph of the 2004 Venus transit shows our nearest planetary neighbor as a dark circle close to the edge of the Sun's disk. This was the first Venus transit since 1882. Another occurred in 2012, but no more are expected until 2117.

PLANETARY CONJUNCTION, APRIL 2002
The conjunction shown here, involving all five naked-eye planets, was visible after sunset for several evenings in April 2002. Although the planets appear close, they are separated by tens or hundreds of millions of miles.

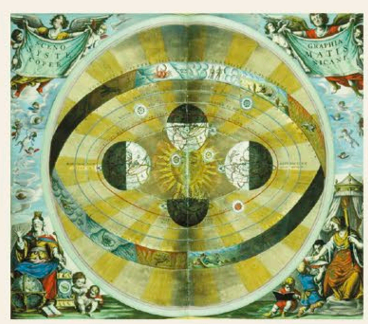
OCCULTATION OF JUPITER BY THE MOON
This occultation occurred on January 26, 2002, and was visible above a latitude of 55°N. Here, the planet sinks out of sight beyond the dark far wall of the lunar crater Bailly. Occultations by the Moon tend to run in series, when for a period the planet and Moon wander into alignment as seen from Earth. An occultation then occurs approximately every sidereal month, until eventually the planet and Moon drift out of alignment again.

NICOLAUS COPERNICUS

Born in Torun, Poland, Copernicus (1473–1543) studied theology, law, and medicine at university. In 1503, he became the canon of Frauenberg Cathedral. This post provided financial security and left him plenty of time to indulge his passion for astronomy. He described his idea of a Sun-centered universe in his book *On the Revolution of the Heavenly Spheres*, published in the year of his death.



At first, Copernicus's revolutionary new idea made little impact. It was only after the telescopic observations of Galileo Galilei and the discovery of the laws of planetary motion by Johannes Kepler (see panel, opposite) that it was finally accepted.



COPERNICAN MAP
This map made by Andreas Cellarius demonstrates the Copernican theory of Earth and the other planets circling the Sun, with the zodiac stars beyond.